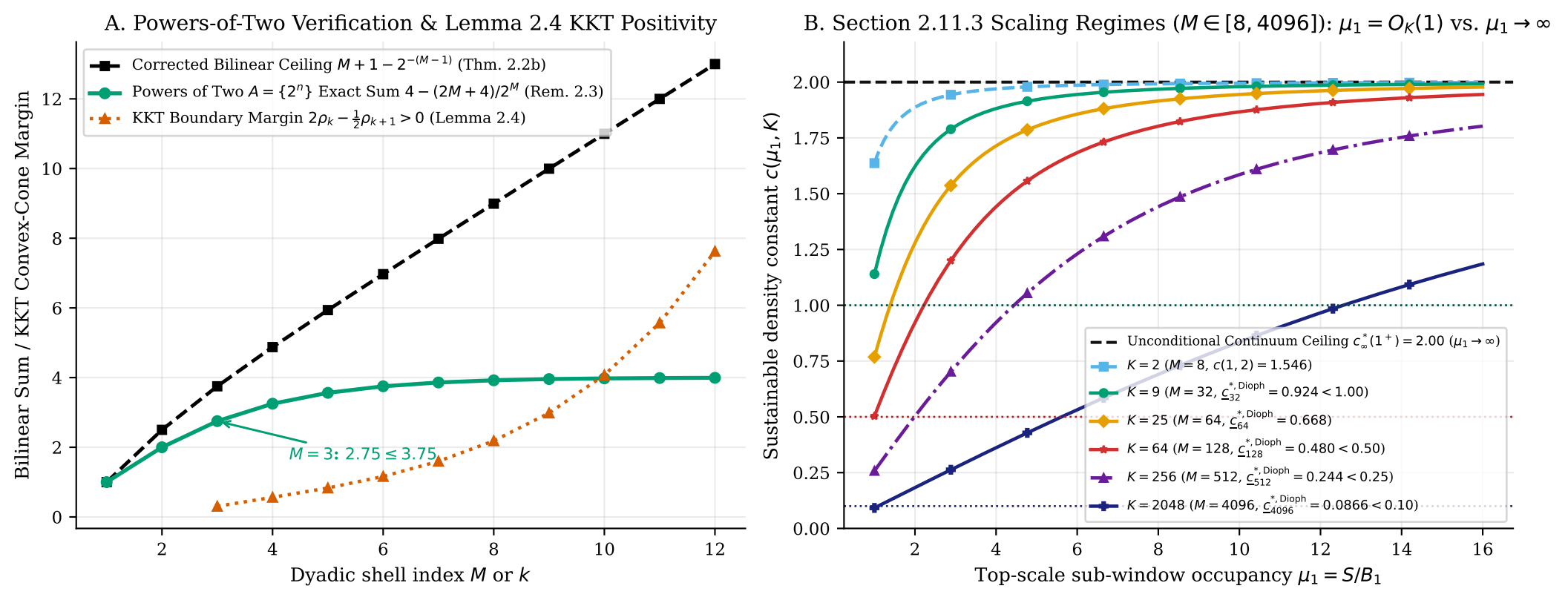
A. Powers-of-Two Verification & Lemma 2.4 KKT Positivity
Corrected Bilinear Ceiling $M+1-2^{-(M-1)}$ (Thm. 2.2b)
Powers of Two $A=\{2^n\}$ Exact Sum $4-(2M+4)/2^M$ (Rem. 2.3)
KKT Boundary Margin $2\rho_k-\frac{1}{2}\rho_{k+1}>0$ (Lemma 2.4)
$M=3$: $2.75\le 3.75$
Bilinear Sum / KKT Convex-Cone Margin
Dyadic shell index M or k
B. Section 2.11.3 Scaling Regimes ($M\in[8,4096]$): $\mu_1=O_K(1)$ vs. $\mu_1\to\infty$
Unconditional Continuum Ceiling $c_\infty^*(1^+)=2.00$ ($\mu_1\to\infty$)
$K=2$ ($M=8$, $c(1,2)=1.546$)
$K=9$ ($M=32$, $\underline{c}_{32}^{*,\mathrm{Dioph}}=0.924<1.00$)
$K=25$ ($M=64$, $\underline{c}_{64}^{*,\mathrm{Dioph}}=0.668$)
$K=64$ ($M=128$, $\underline{c}_{128}^{*,\mathrm{Dioph}}=0.480<0.50$)
$K=256$ ($M=512$, $\underline{c}_{512}^{*,\mathrm{Dioph}}=0.244<0.25$)
$K=2048$ ($M=4096$, $\underline{c}_{4096}^{*,\mathrm{Dioph}}=0.0866<0.10$)
Sustainable density constant $c(\mu_1,K)$
Top-scale sub-window occupancy $\mu_1=S/B_1$